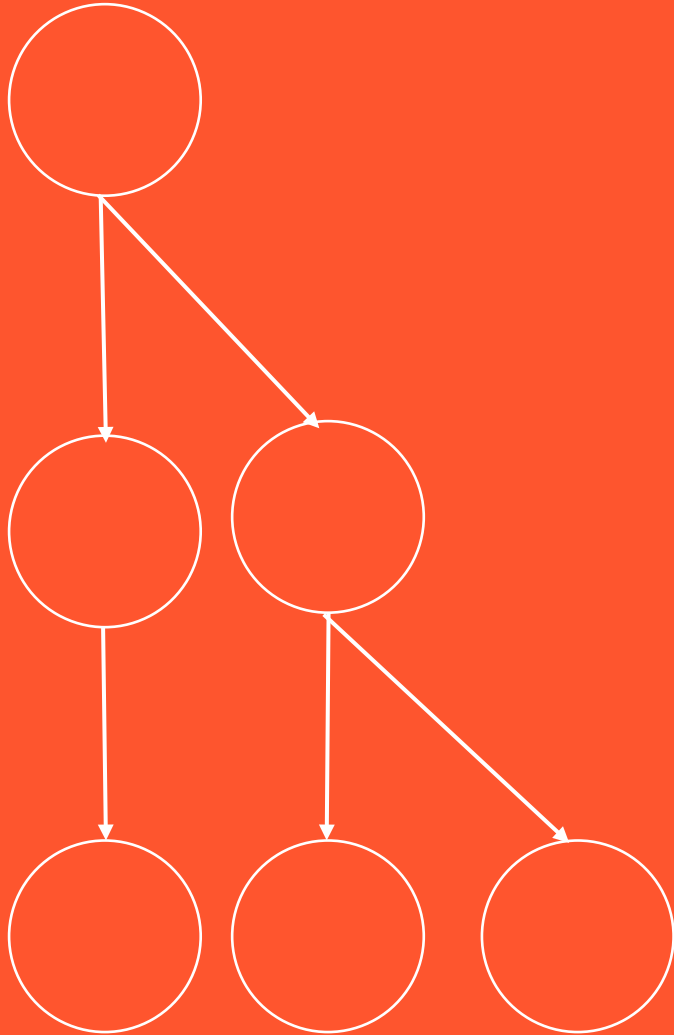


BCOG 100

Symbolic Cognition

Lecture 2: The Brain and Cognitive Science of Art, Music, and
Storytelling



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 12-2

Password: LoT

Question: What is the Language of Thought hypothesis?

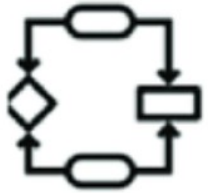


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Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

Last Time:

The Symbolic Mind and the Cultural Explosion

Today:

Language made meaning shareable

Now symbols become expressive

Art, music, and story extend the mind

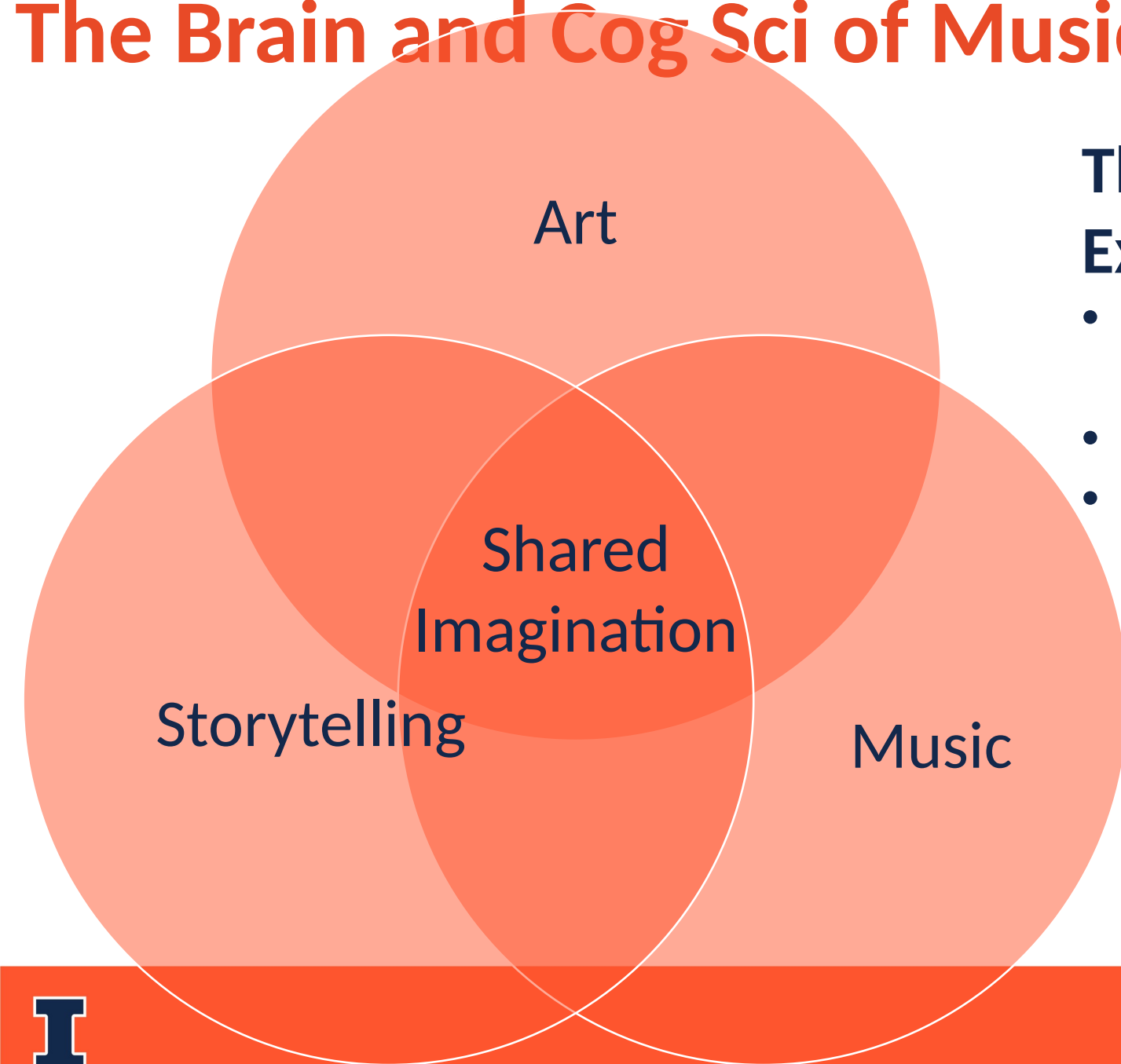
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What Makes Expression Special?

- A puzzle for evolution
- Art, music, and story cost time and energy
- Yet every culture makes them

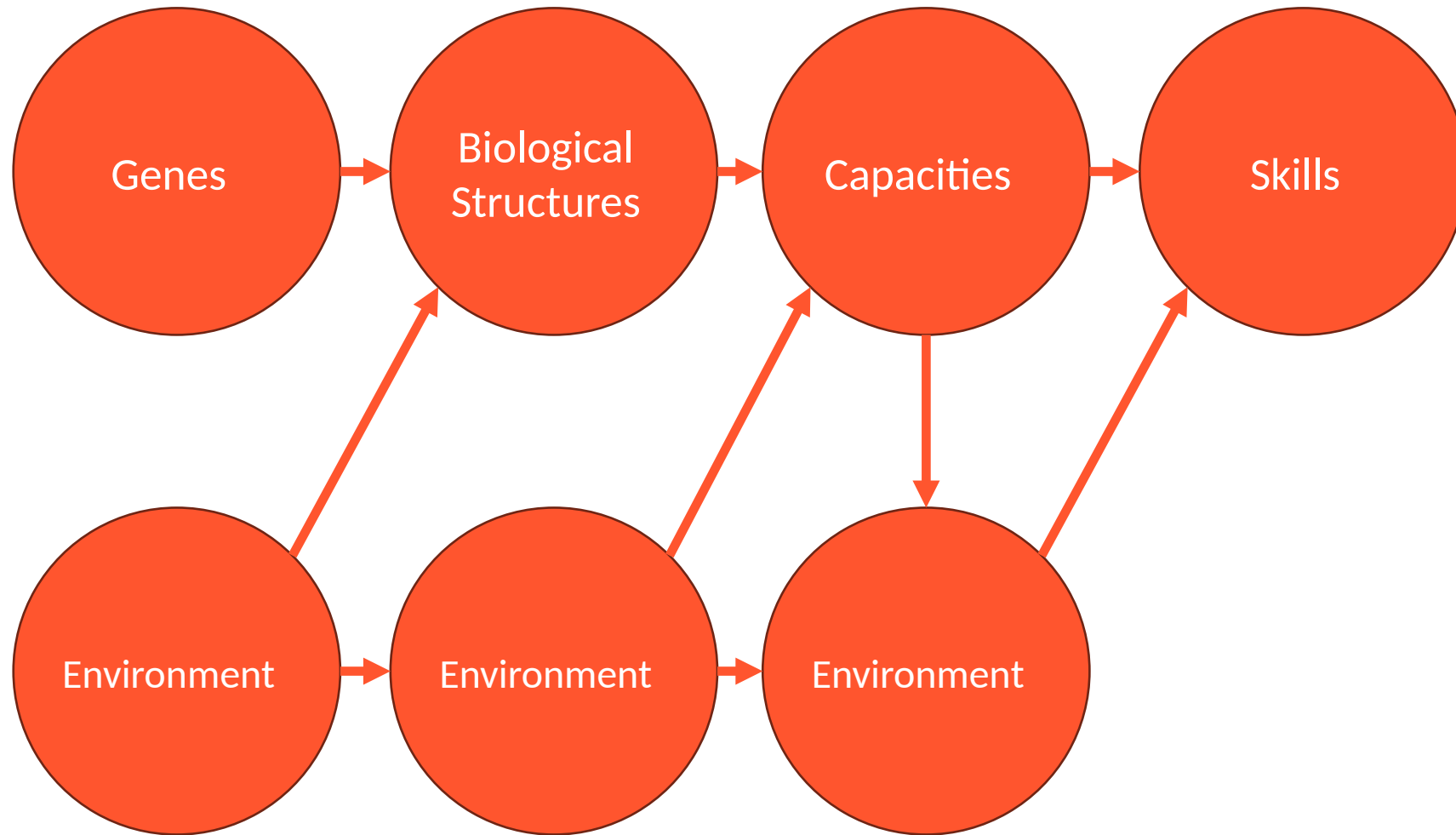
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Three Windows into the Expressive Mind

- Music, story, and art as shared imagination
- Each extends older cognitive systems
- Together, they define what culture is

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From Genes to Skills:

Levels and Loops

- Genes set constraints, not skills
- Brains bias learning, culture builds
- Practice sculpts expertise

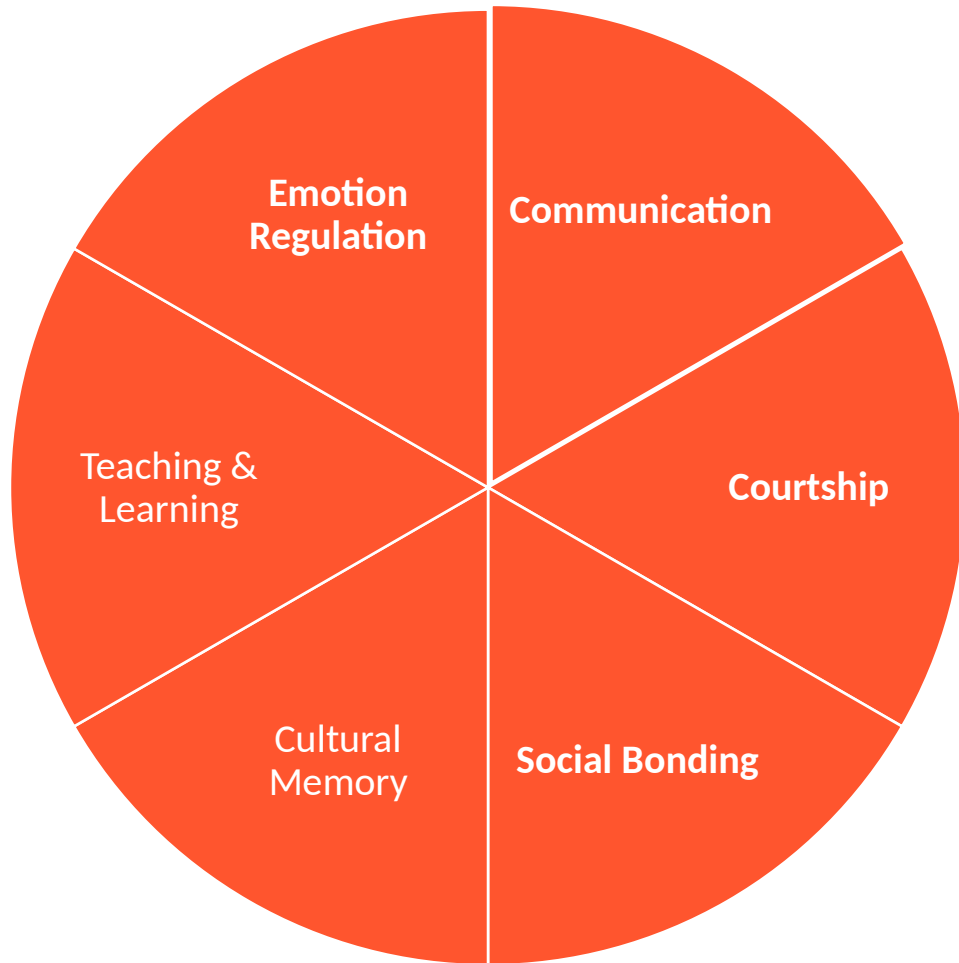
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The Origins of Rhythm and Song

- Rhythm before language
- Emotion before meaning
- Music as ancient social glue

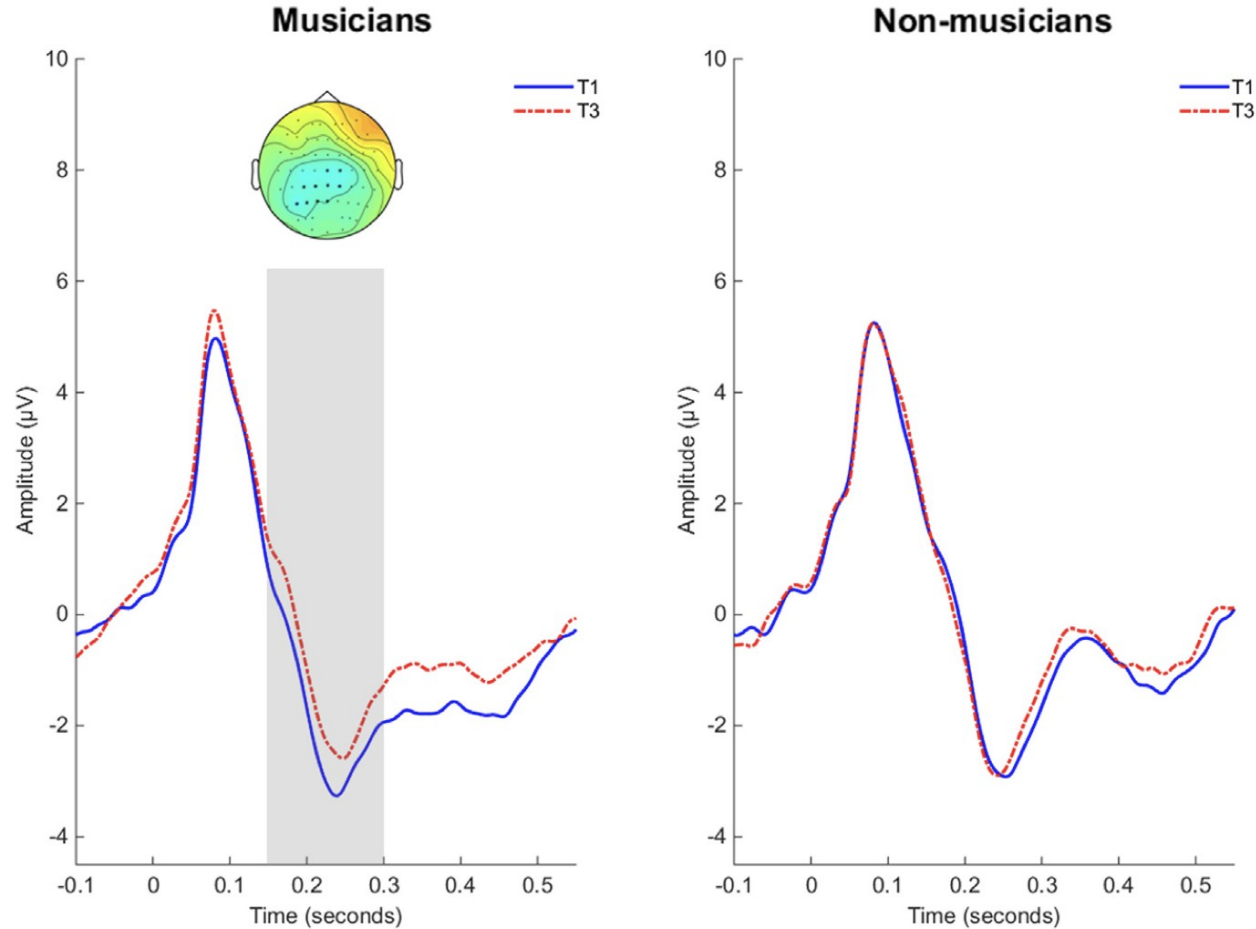
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Why Did Music Evolve?

- An evolutionary puzzle with many answers
- Bonding, signaling, and synchrony
- Music as a tool for social prediction

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Music as Predictive Pattern Learning

- Brains predict sound
- Pleasure follows accurate prediction
- Music exploits that loop

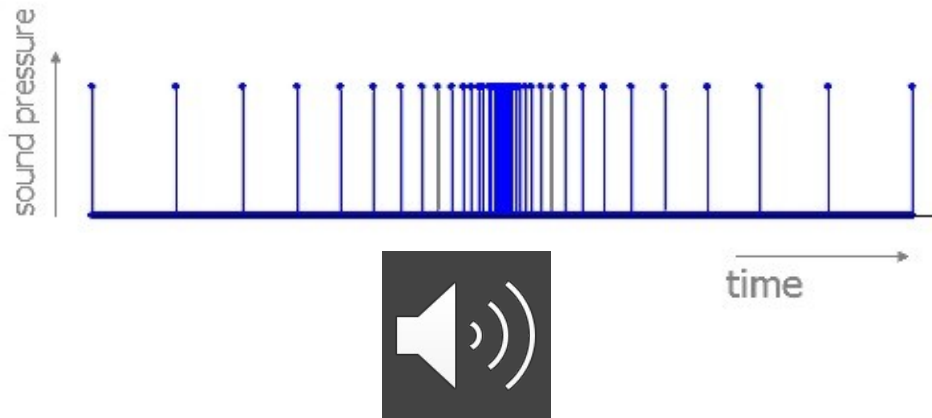
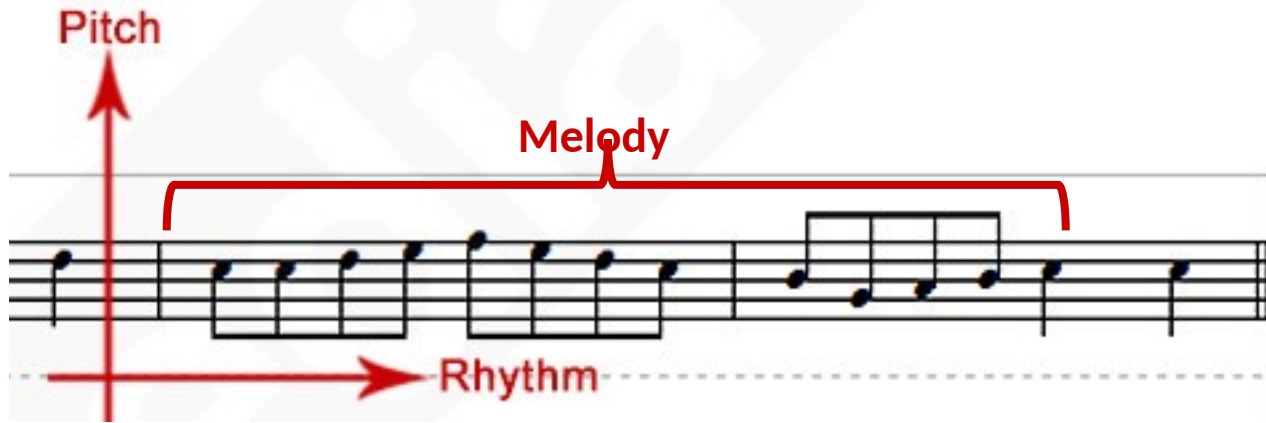
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Musical Beginnings: Infants and Early Sensitivity

- Music before language
- Prediction before instruction
- Emotion and rhythm from the start

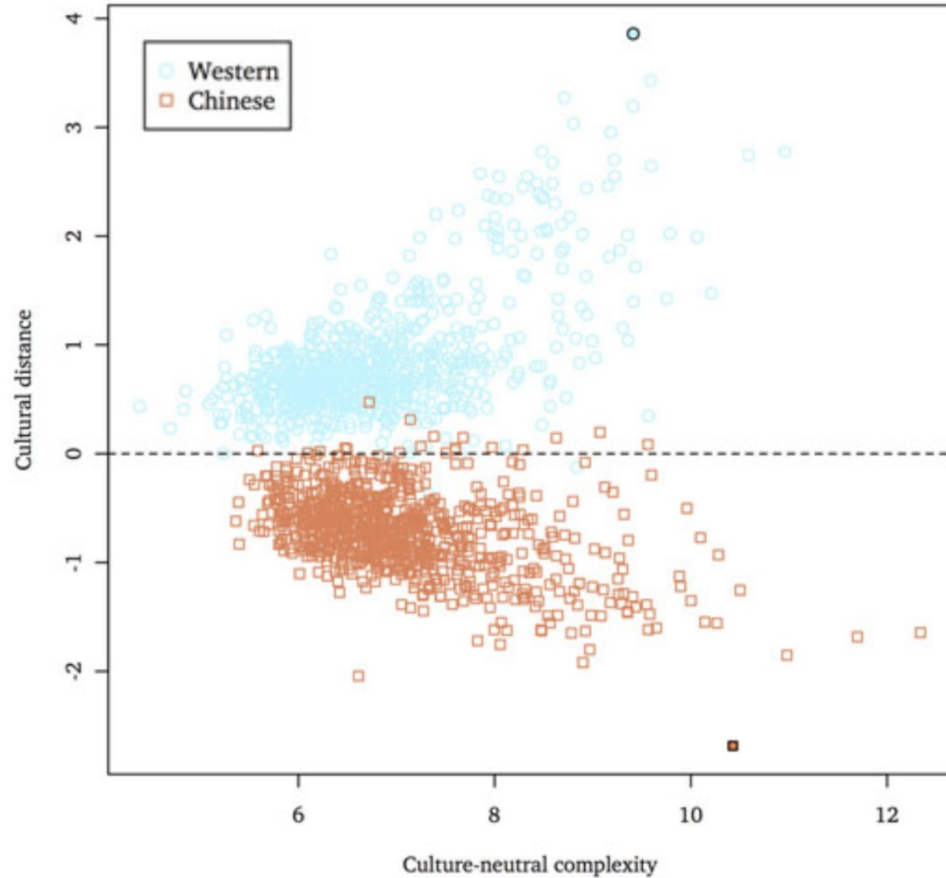
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Rhythm and Pitch in Music Cognition

- Time and tone as twin hierarchies
- Prediction and schema guide perception
- Shared structure with language and movement

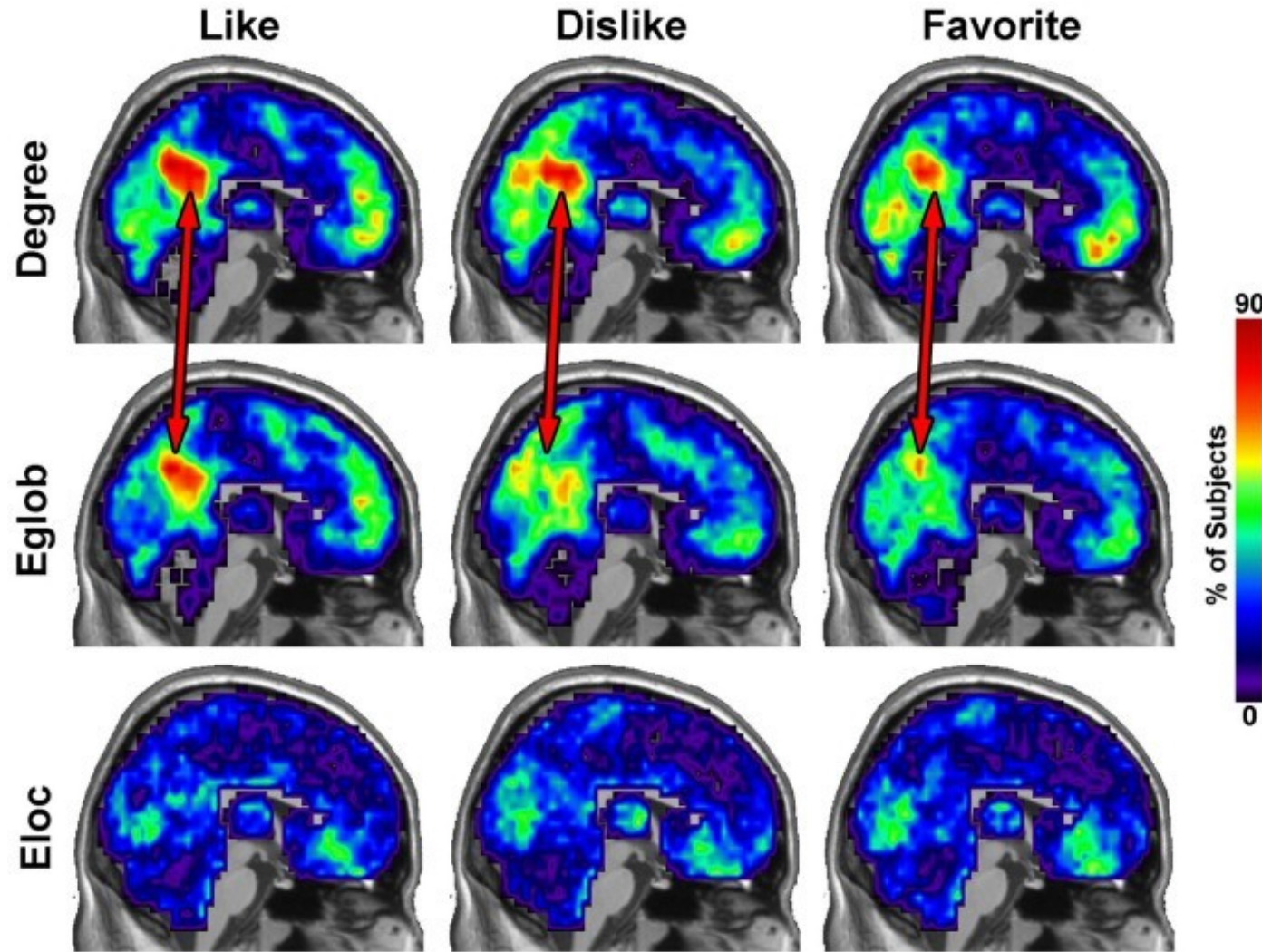
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Musical Enculturation as Predictive Learning

- Music as learned prediction
- Culture shapes statistical models
- Enculturation tunes expectation and emotion

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The Musical Brain

- Like other faculties, no single “music center”
- Distributed networks for sound, action, and reward
- Prediction connects them all

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From Song to Story

- Rhythm became sequence
- Sound became meaning
- Shared feeling became shared thought

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The Evolutionary Function of Storytelling

- Simulating without risk
- Teaching through imagination
- Building “we” through shared worlds

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Cognitive Foundations of Narrative

- Story as language plus simulation
- Simulating minds through words
- Culture aligns what we imagine

Story Elements	Cognitive Functions
Setup / Exposition	Establishes context, agents, and goals; activates prior knowledge and schemas.
Conflict / Problem	Violates expectation; recruits prediction-error and attention systems to detect change.
Rising Action	Generates hypotheses about causes and outcomes; engages working memory and simulation.
Climax	Tests predictions and emotional investment; integrates surprise, tension, and resolution processes.
Resolution	Updates mental models; reinforces causal learning and emotional regulation.

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Narrative Structure and Predictive Models

- Stories as prediction engines
- Conflict as violated expectation
- Resolution as learning and reward

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Emotion, Empathy, and Identification

- Stories as emotional simulations
- Feeling others' feelings
- Learning who we are through who we imagine

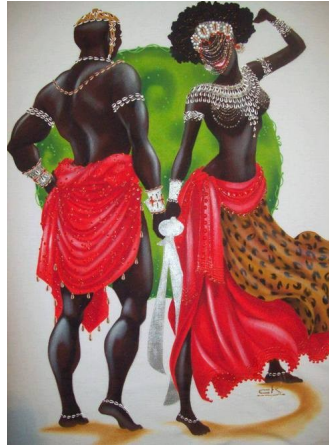
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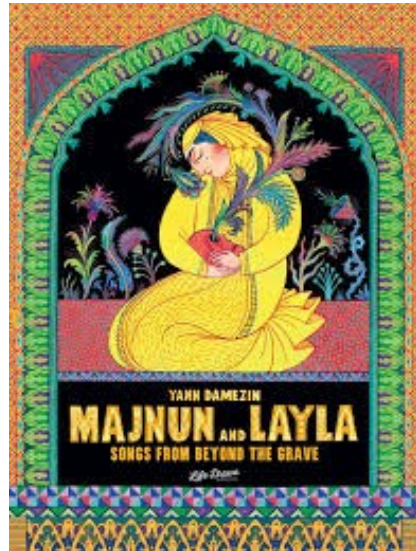
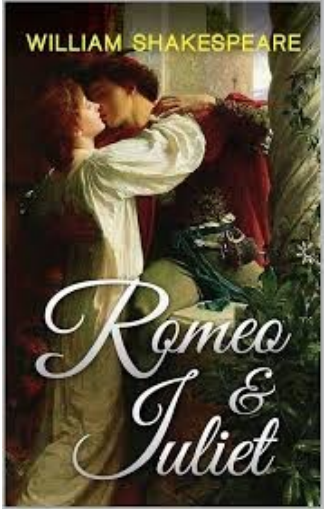
The Butterfly Lovers



Heer Ranjha



Oya and Shango



Xquic and Hun Hunahpu

The Cultural Evolution of Storytelling

- From myth to novel
- From oral to written to digital
- Stories evolve as minds and media do

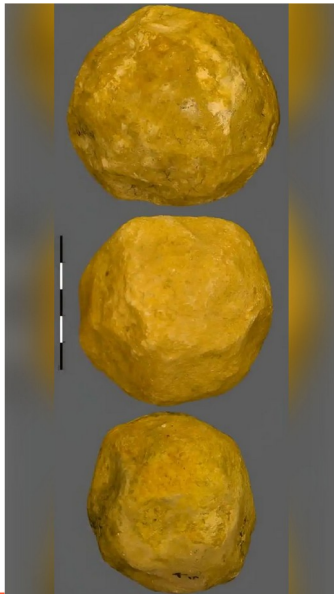
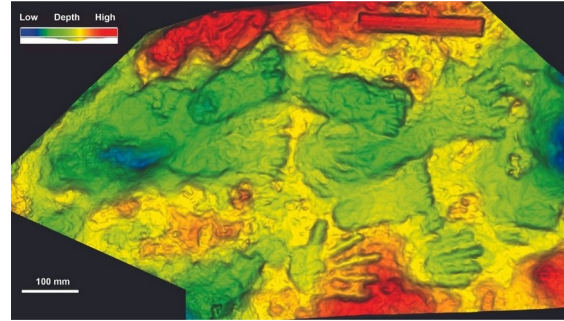
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Why Art? The Puzzle of Aesthetic Behavior

- Beauty without utility
- Effort without survival gain
- Yet found in every culture

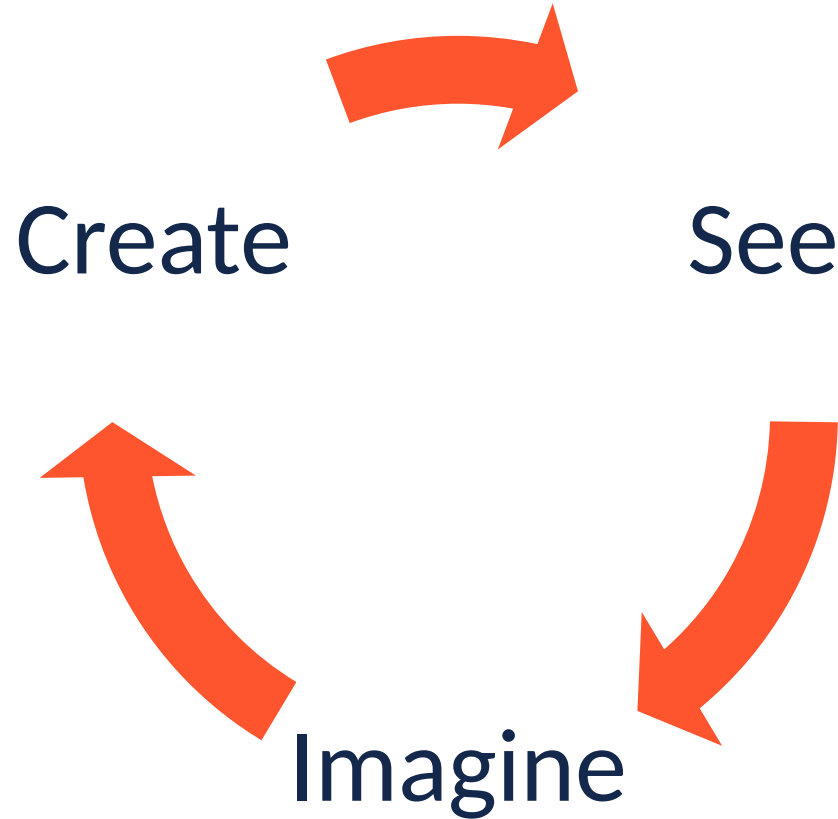
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The Evolutionary Origins of Image-Making

- Hands before brushes
- Symbols before scenes
- Art as the mirror of imagination

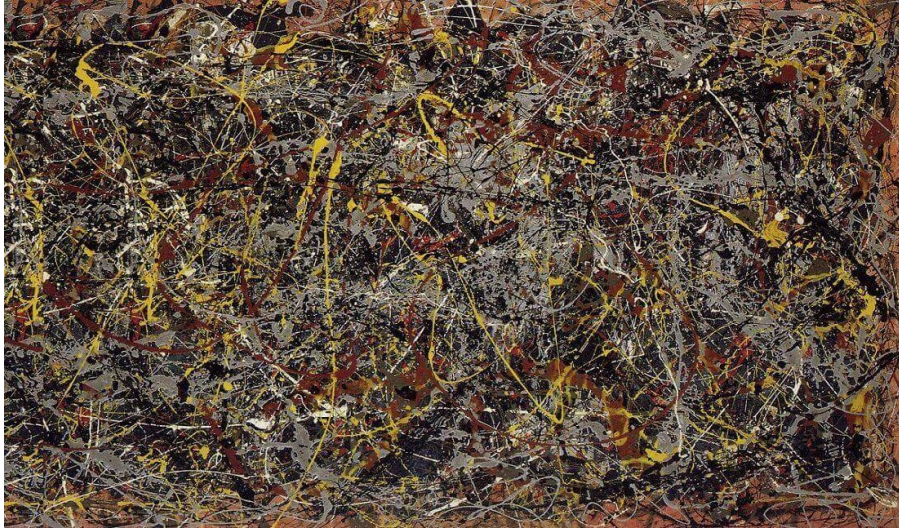
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The Cognitive Foundations of Art

- Seeing as imagining
- Patterns as predictions
- Perception and creation share a code

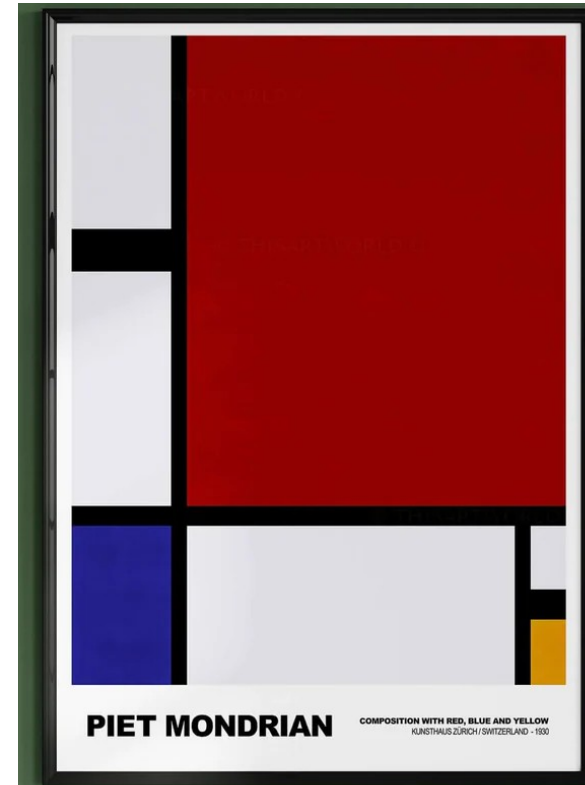
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Jackson Pollock, No. 5



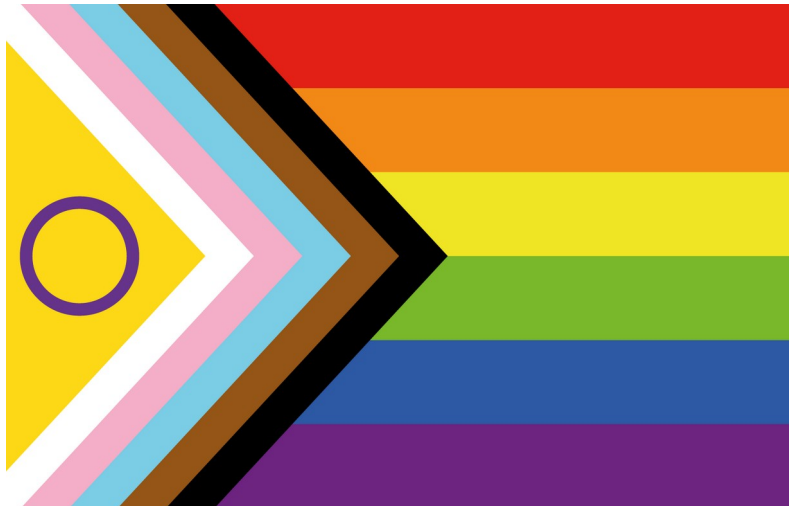
Claude Monet , San Giorgio at Dusk



Perception, Prediction, and Aesthetic Pleasure

- The brain loves to predict
- Beauty lives between order and surprise
- Art rewards accurate imagination

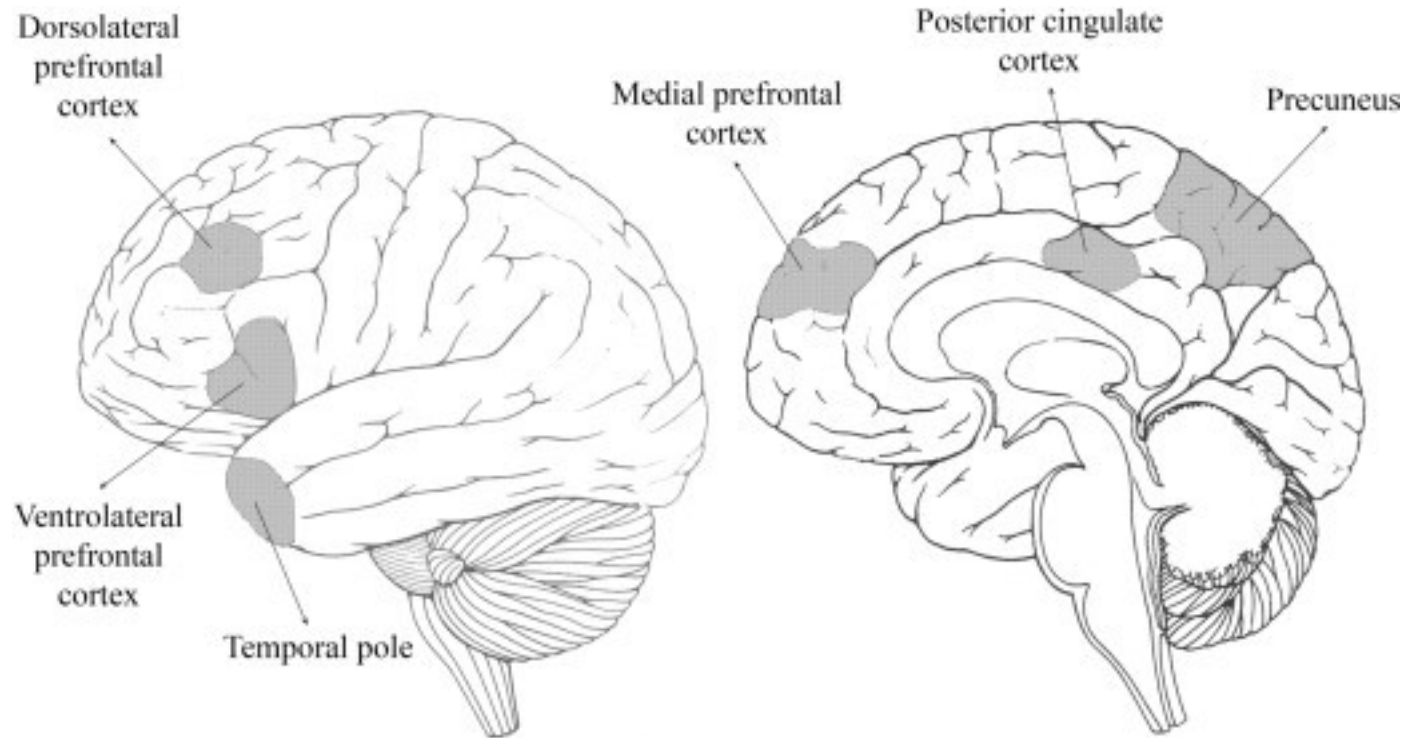
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Art, Symbol, and Culture

- Images as shared memory
- Symbols as social contracts
- Art as collective imagination made visible

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The Artistic Brain

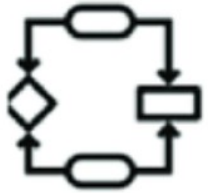
- Vision meets imagination
- Action meets emotion
- Art as perception turned inside out

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Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Summary

- Expression as shared imagination
- From sound to story to image
- Prediction and pleasure in every art
- Culture as the mind made visible

Next Time

- Writing, Mathematics, and Science